

Free radicals & Antioxidants

Prepared by

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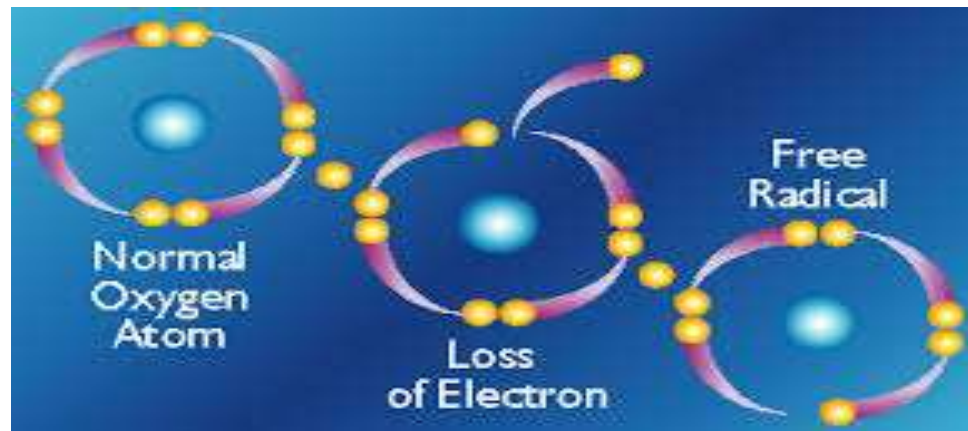
MSc. PhD. In Clinical Biochemistry/Medicine College/HMU

1st Stage

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Definition:

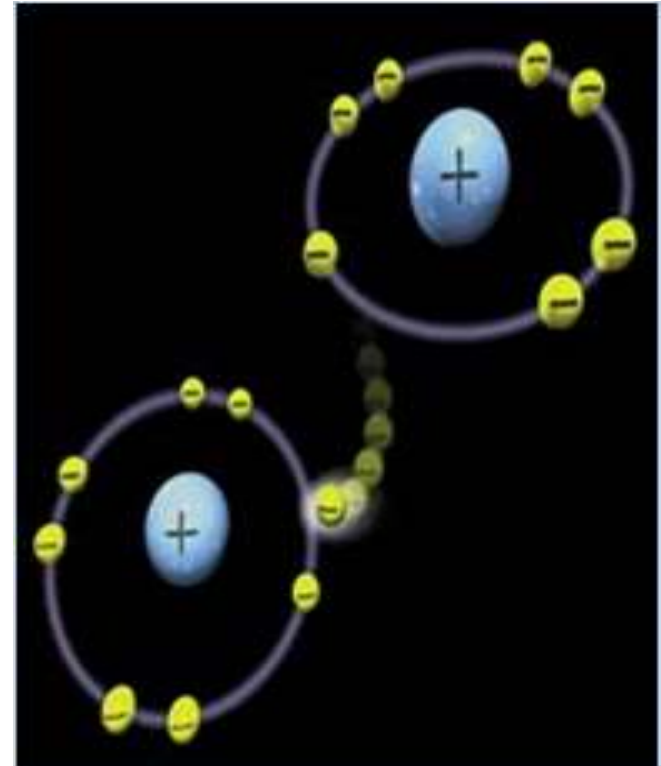
- A free radical is an atom, molecule, or ion that contains **one or more unpaired electrons** in its outer shell.
- Free radicals may have positive, negative, or zero charge.
- It's represented by a **superscript dot** to the right ($R^\bullet$).



Important characteristics:

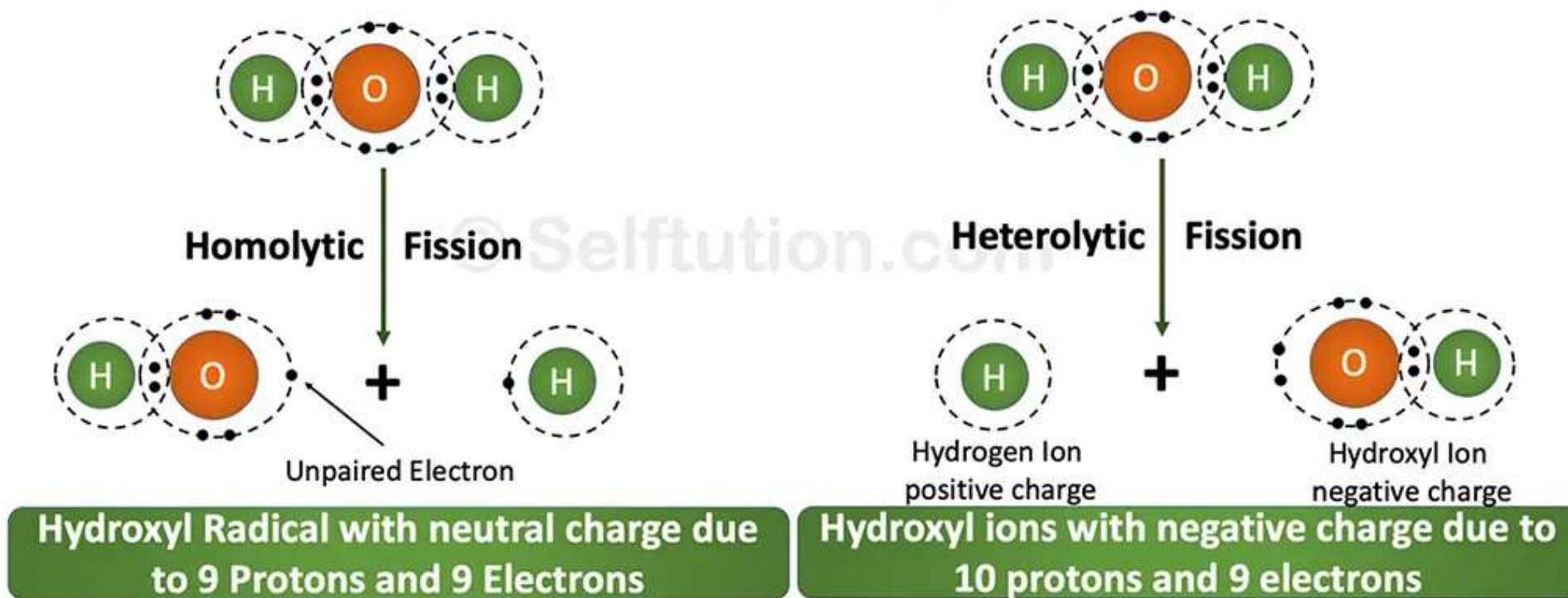
❑ Free radicals are:

- Highly reactive
- Unstable and try to become stable.
- Short life span as they tend to catch an electron from other molecules.



How radicals can be formed?

Homolytic and Heterolytic Fission of Water Molecule



Differences between Free radicals and Ions

Free radicals

- A radical is a molecule with one or more unpaired electrons, making it highly reactive.
- Mostly neutral
- Less stable
- Often formed by breaking apart molecules during chemical reactions, exposure to light or heat, or by other radicals.

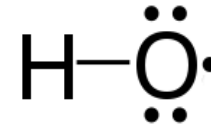
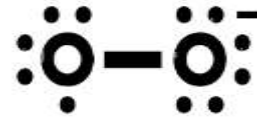
Ions

- An ion is an atom or molecule that has gained or lost electrons, resulting in a net electrical charge.
- Charged (either as positive or negative).
- Generally more stable
- Formed by gaining or losing electrons due to factors like ionization energy (energy needed to remove an electron), electron affinity (attraction for electrons), or chemical reactions.

Types of Free Radicals:

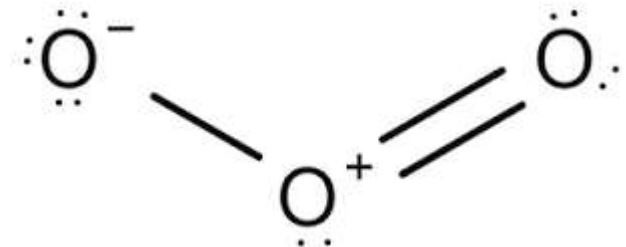
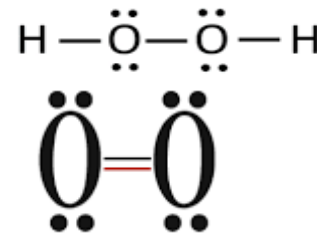
□ Oxygen free radicals(OFR):

- Superoxide radical($\text{O}_2^{\bullet-}$)
- Hydroxyl radical($\text{OH}^\bullet$)



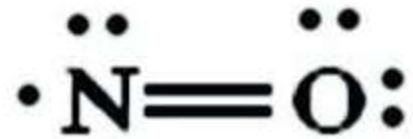
* **Reactive oxygen species(ROS)**, includes both (OFR) and non radicals that are oxidizing agents and/or easily converted into radicals such as:

- Hydrogen peroxide(H_2O_2)
- Singlet oxygen($^1\text{O}_2$)
- Ozone O_3



❑ Nitrogen free radicals(NFR):

- nitric oxide radical NO•
- peroxynitrite anion ONOO-
- nitrite anion NO₂-



❑ Lipid free radicals(LFR):

- L•, LO•, and LOO•

❑ **Others**, chlorine radicals(Cl•), carbon radicals (CCl₃•), and sulfur radicals(RS•).

Sources of Free Radicals

Endogenous sources of FR

* Free radicals are generated as natural by-products of metabolic processes:

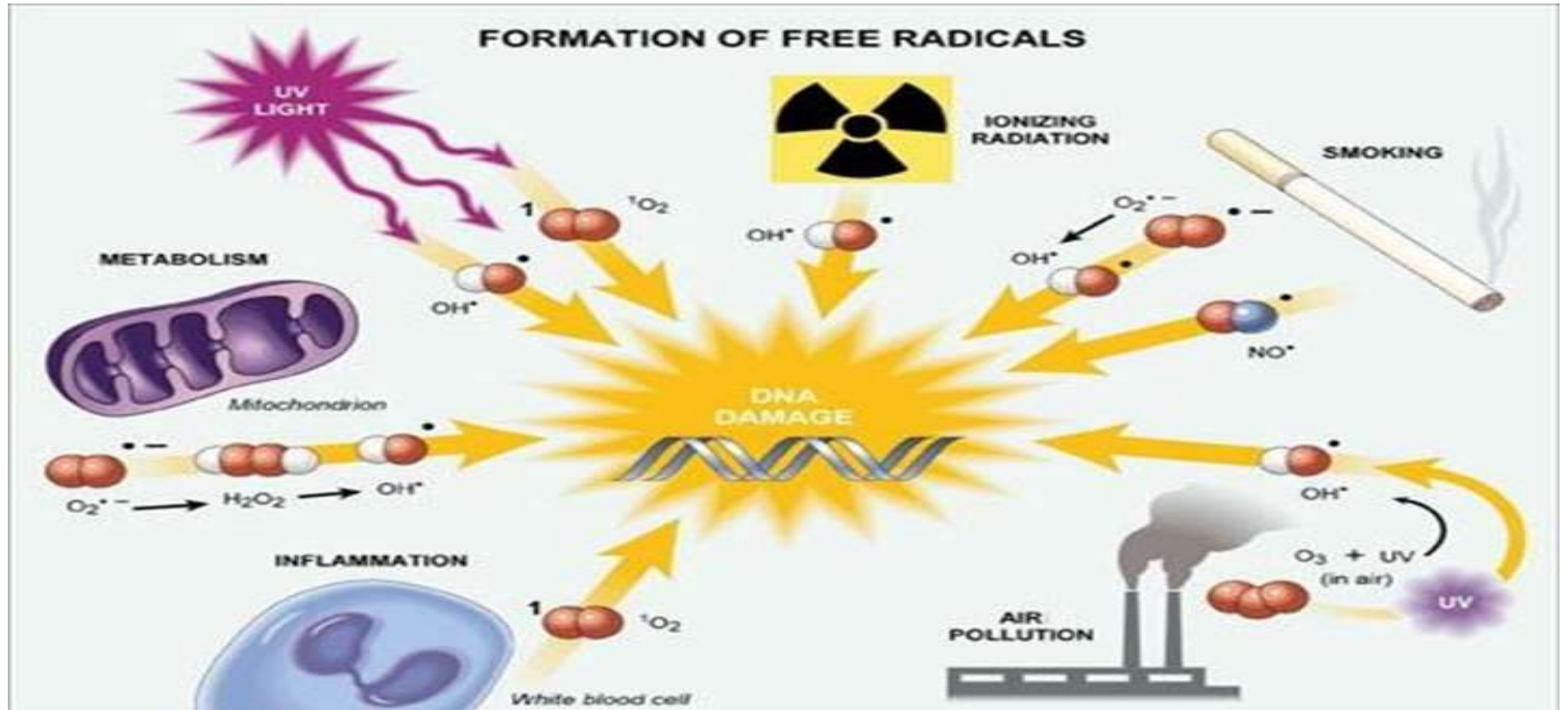
- **Mitochondrial Respiration**
- **Metabolism**
- **Inflammation & immune response**
- **Phagocytosis**

Exogenous sources of FR

* They arise from exposure to environmental factors:

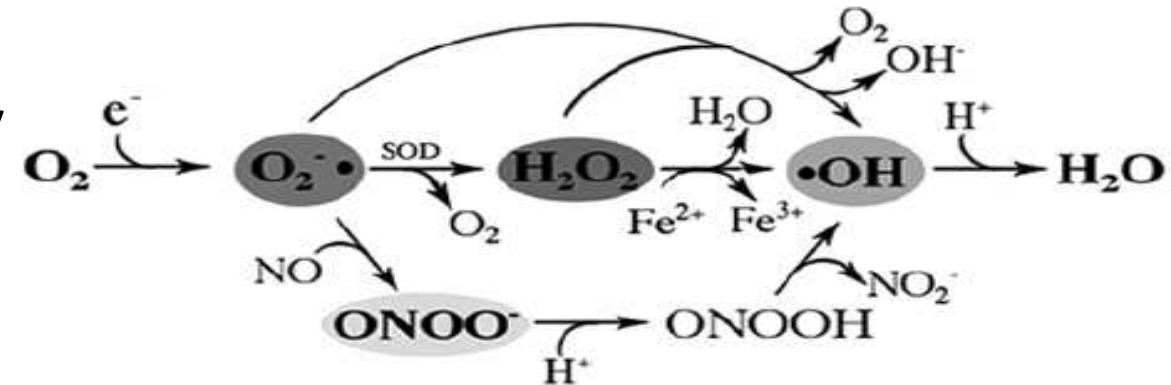
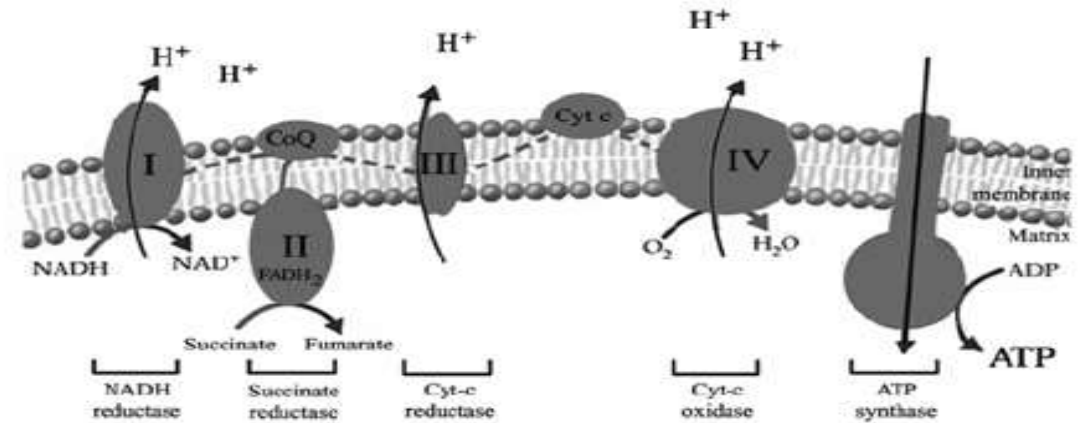
- **Ionizing radiation (x-ray & gamma ray)**
- **Ultraviolet radiation**
- **Chemicals, smoking**
- **Air pollution**
- **Dietary sources (oxidized fats, food additive).**

Endogenous & Exogenous FRs



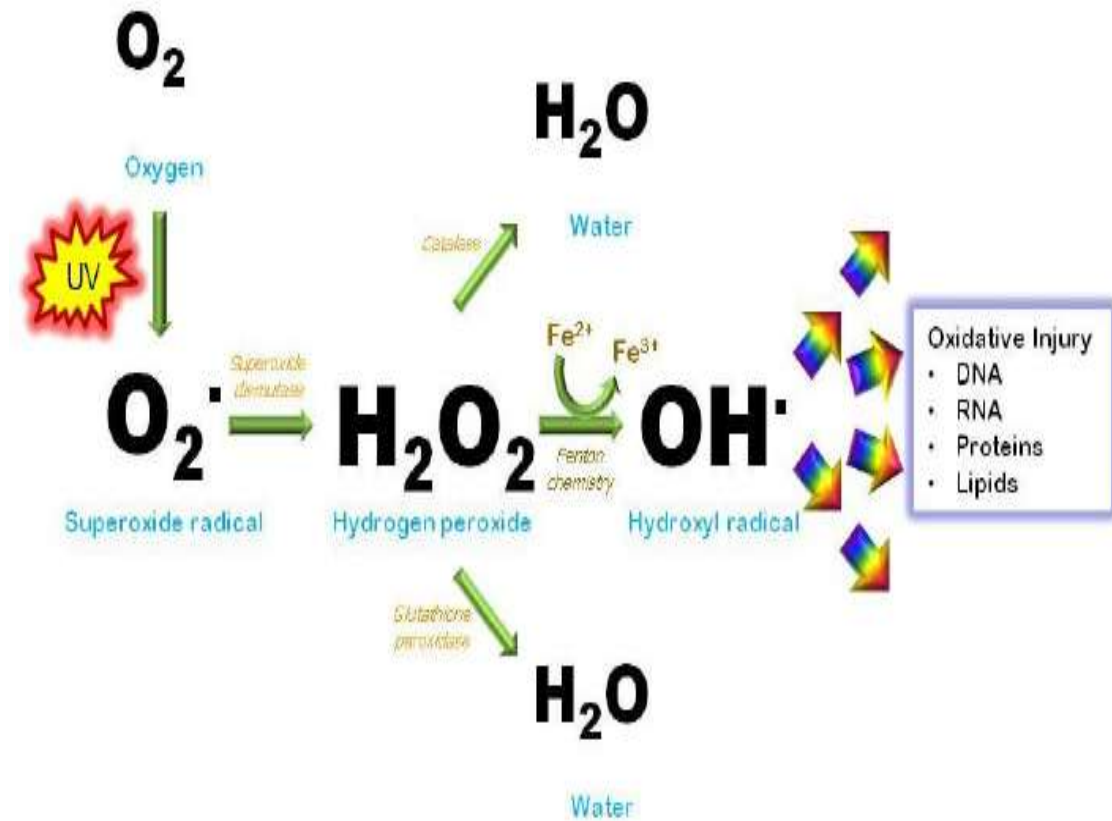
Mechanism of FR generation from Respiration:

- In **mitochondrial respiratory chain**, electron transfer through Complexes I, III, and IV generates a **proton gradient** that is used to generate **ATP** through **ATP synthase** (Complex V).
- During respiration some electrons go directly to oxygen forming **superoxide anion** ($\bullet\text{O}^{2-}$) that leads to the production of several other FRs.



Mechanism of FR generation from UV light:

- UV light generates ROS primarily through **photosensitization**.
- UV light is absorbed by endogenous photosensitizers like (porphyrins, flavins, melanin, NADH).
- These excited molecules interact with molecular oxygen or other organic molecules to produce highly reactive radical species like ($O_2^{\bullet-}$, H_2O_2 , $OH^{\bullet}$).
- This process, induced by **UVA** and **UVB**, triggers **oxidative stress**, **DNA damage**, and **inflammatory pathways**.



Roles of Free Radicals in Biological Systems

- Low to moderate levels of free radicals play an important role in a number of biological processes for the human health, the most important are:
 1. The **immune system's cells** (phagocytes), produce oxygen free radicals to destroy invading bacteria, viruses, and other pathogens.
 2. The free radical **nitric oxide** ($\cdot\text{NO}$) is a vital messenger molecule that helps to relax and dilate blood vessels, so **reduce blood pressure**.

3. FRs help to remove damaged or cancerous cells, so assist in **programmed cell death (apoptosis)**.
4. Act as **signaling molecules**, helping regulate cell growth and differentiation, and gene expression.
5. Modify **oxidation-reduction** states.

Mechanism of radical reactions

➤ Three distinct steps:

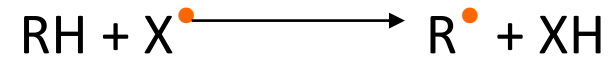
i- **Initiation**(homolytic covalent bonds cleavage)

ii- **Propagation**(chain propagation)

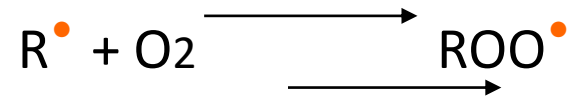
iii- **Termination**

Oxidative damage to lipids- Lipid peroxidation

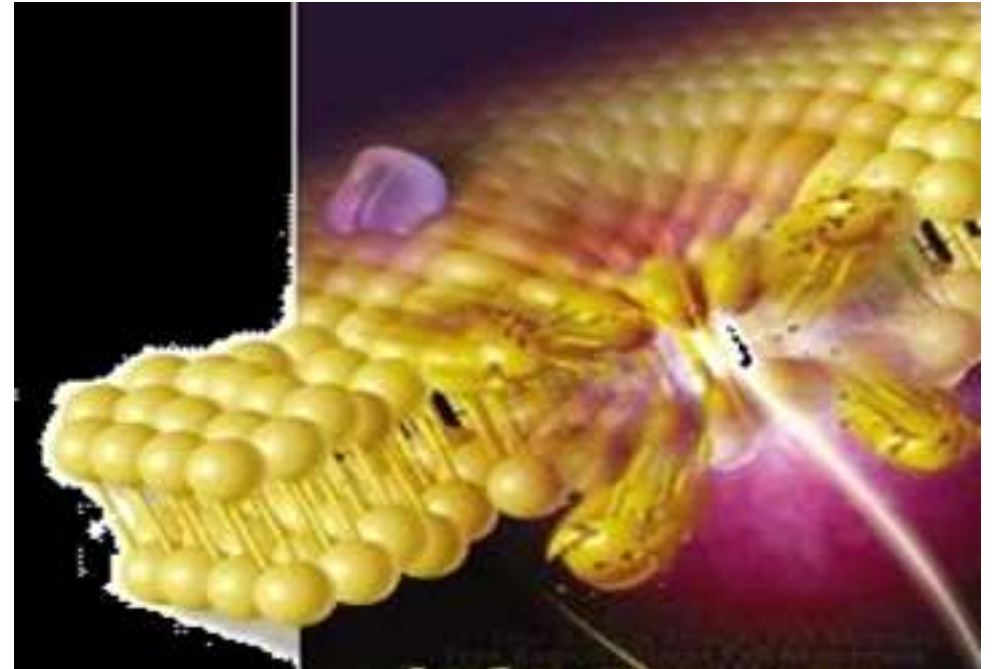
i- Initiation:



ii- Propagation:



iii- Termination:

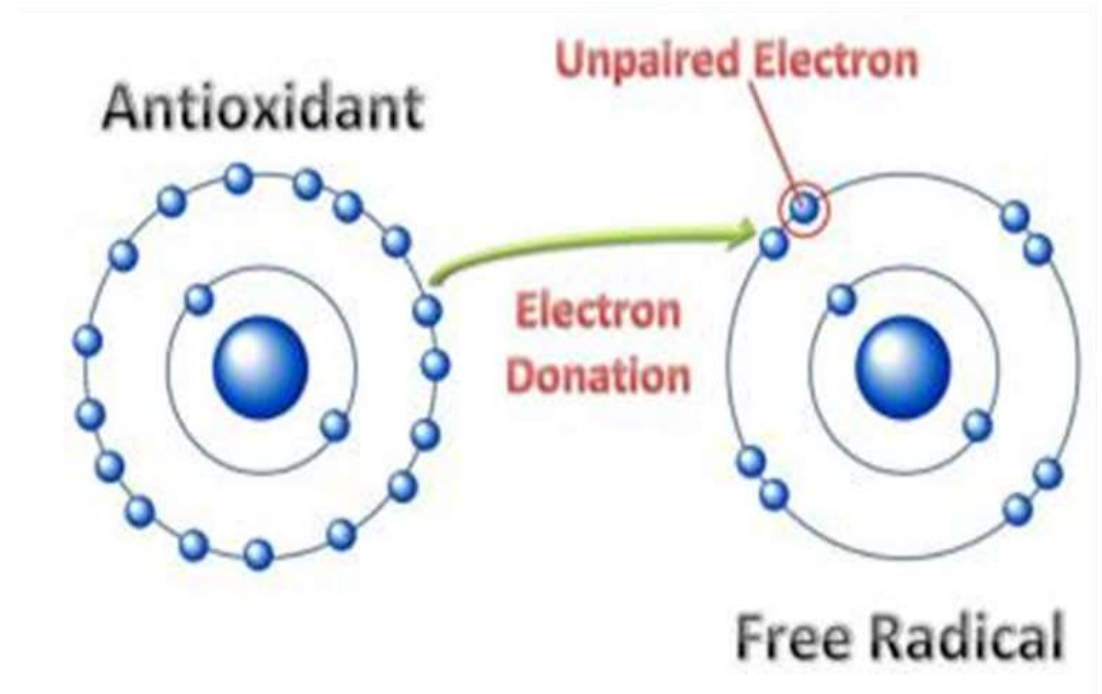


Antioxidants

- The substance present in low concentrations relative to the oxidizable substrate that significantly delays or reduces oxidation of the substrate.
- They reduce the effect of dangerous oxidants by binding together with these harmful molecules, decreasing their destructive power.
- They can also help repair damage already sustained by cells.
- They may be considered as the scavengers of free radicals.
- During this reaction the antioxidant sacrifices itself by becoming oxidized.

Antioxidants

- Stabilizes free radicals.
- Terminates free radical reactions.
- Prevents the transfer of electron from O_2 to organic molecules.



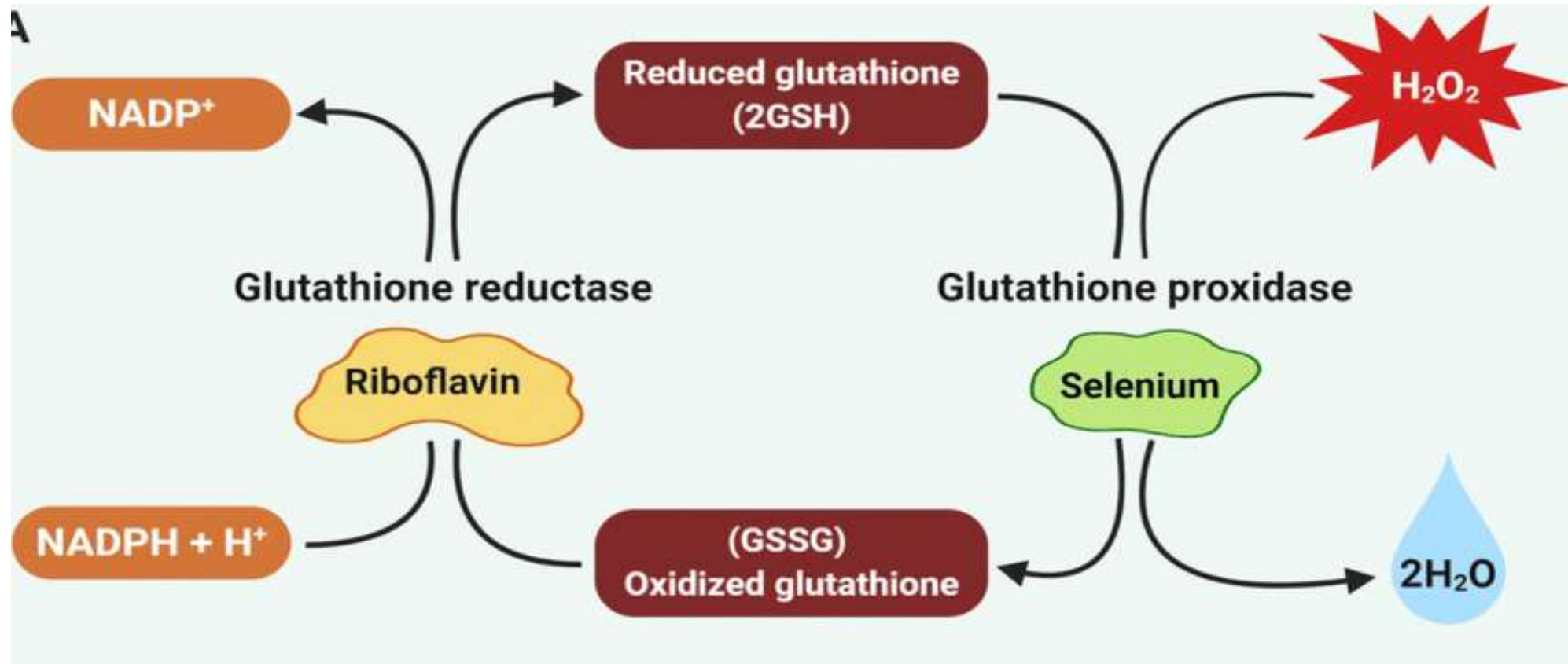
Classification of Antioxidants

According to their nature and action:

1- Enzymatic antioxidants (made by the body):

- Superoxide dismutase (SOD), converts superoxide into H_2O_2
- Catalase, converts H_2O_2 into H_2O and O_2
- Glutathione Peroxidase (GPx)
- Glutathione reductase (GR)

Glutathione antioxidant system



2- Non-enzymatic antioxidants:

- **Nutrient antioxidants (from diet):**

Vitamin A/ β -carotene, Vitamin E, Vitamin C, Vitamin K, Glutathione (GSH), Polyphenols (tea, fruit), and Selenium (cofactor for GPx).

- **Metabolic antioxidants (made in body):**

bilirubin, uric acid, ceruloplasmin, ferritin, transferrin, albumin, glutathione (GSH), and so on.

8 Foods Rich in Antioxidants



Berries



Plums



Bell Peppers



Green Tea



Spinach



Kale



Broccoli



Nuts

What is oxidative stress?

- Oxidative stress is an **imbalance** between the **production of free radicals** and the ability of the body to detoxify their harmful effect through neutralization by **antioxidants**.
- The excess free radicals causes **damage** to **lipids, proteins, and DNA**, which can lead to **cell death** and **tissue damage**.
- Oxidative stress leads to many **pathological conditions**, some of these are the following:

- 1. Cancer:** oxidative stress **damages the DNA** in healthy cells, which can raise the risk of cancer.
- 2. Cardiovascular disease:** oxidative stress triggers **plaque** to form in the arteries (atherosclerosis). Plaque buildup can lead to many different problems, including **reduced blood supply** to the heart (coronary artery disease) and heart attack.
- 3. Kidney disease:** sustained oxidative stress can form **scar tissue** in the kidneys that prevents them from working, this may lead to kidney failure.

4. Neurological diseases: oxidative stress leads to different neurological diseases, including Alzheimer's disease, Parkinson's disease and multiple sclerosis. Excess free radicals contribute to **neuron loss**, which can progressively lead to dementia.

5. Respiratory diseases: oxidative stress and related inflammation can affect the **lungs** and cause conditions like **asthma** or **chronic obstructive pulmonary disease (COPD)**.

- 6. Rheumatoid arthritis:** free radicals contribute to the **chronic inflammation** that happens in people with **rheumatoid arthritis**.
- 7. Diabetes mellitus:** **destruction** of **islet cells** results in pathogenesis.
- 8. Aging Process:** aging results from cumulative **oxidative damage** caused by free radicals to **DNA, proteins, lipids, and mitochondria**, leading to progressive cellular and organ dysfunction.

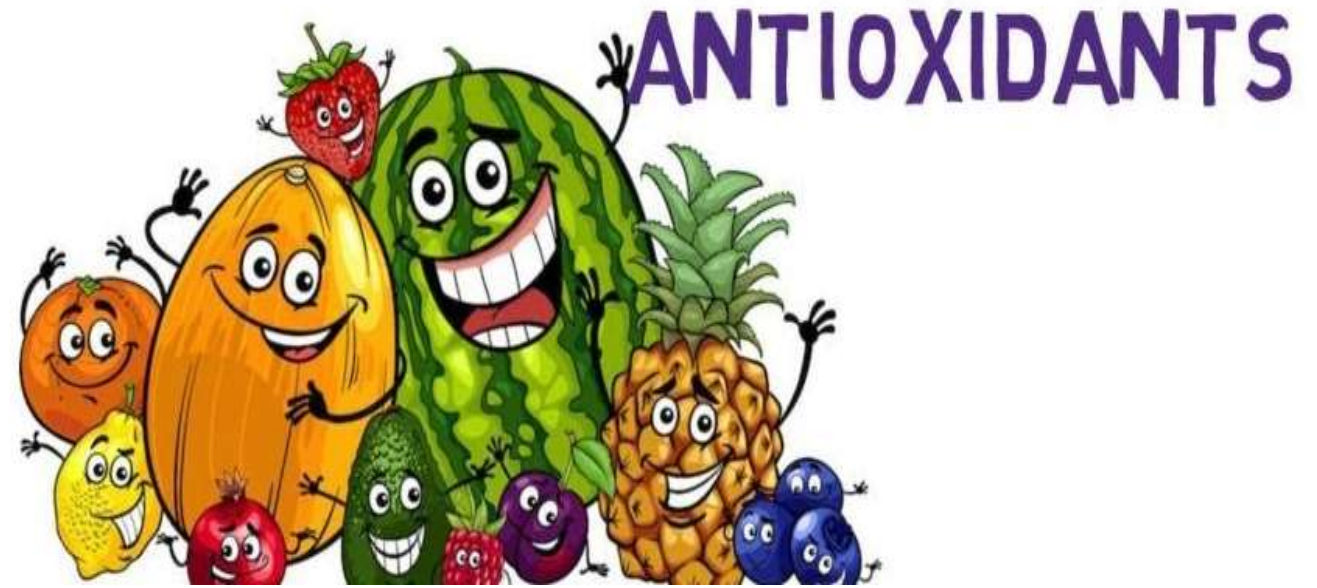
ROS and Antioxidants that Reduce them

Reactive Species	Antioxidant
Singlet oxygen $^1\text{O}_2$	Vitamin A, Vitamin E
Superoxide radical $\text{O}_2^{\bullet-}$	Superoxide dismutase, Vitamin C
Hydrogen peroxide H_2O_2	Catalase, glutathione peroxidase
Peroxyl radical $\text{ROO}^\bullet$	Vitamin C, Vitamin E
Lipid peroxyl radical $\text{LOO}^\bullet$	Vitamin E
Hydroxyl radical $\text{OH}^\bullet$	Vitamin C

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FREE RADICALS

